Arithmetic and complexity

Exact bilinear rank of the 3×3 matrix product

Difficulty: extreme **Importance:** interesting to the community

Status: Open

Rating rationale: Extreme because the exact rank of this small multiplication tensor remains a decades-old barrier despite extensive algorithm searches; community importance concerns recursive matrix multiplication and bilinear algorithm design.
Topic: small matrix multiplication algorithms

Problem statement

Determine the least integer r for which there are complex coefficients $u_{t,ij}, v_{t,ij}, w_{ij,t}$ satisfying, for every pair of complex 3×3 matrices,

$$(AB)_{ij} = \sum_{t=1}^{r} w_{ij,t} \left(\sum_{a,b=1}^{3} u_{t,ab} A_{ab} \right) \left(\sum_{c,d=1}^{3} v_{t,cd} B_{cd} \right).$$

Scalar linear combinations are free in this bilinear-rank model. The classical upper bound is 23; deciding whether 22 products suffice is part of determining the exact minimum. Algorithms mixing entries of both inputs within a factor are outside this specified model.

Why it matters

Such identities can be applied recursively to matrix blocks.

References and status

Bläser, 2013, §1 and Example 5.10. Y. Sun, *An Exact 56-Addition, Rank-23 Scheme for General 3×3 Matrix Multiplication* (2026), gives another 23-product scheme. C. Wang, *Automated Lower Bounds for Bilinear Complexity over Finite Fields*, v10 (2026), improves a lower bound over $\mathbb{F}_2$; that is not a complex-field resolution. Searches for "rank 22 matrix multiplication 2026 proof" found no qualifying algorithm or matching lower bound. **Admitted: no resolution located.**

Status check — 2026-09-10

Rechecked Sun's rank-23 construction and Alman–Li, §1, which explicitly identifies both exact rank and border rank of the 3×3 product as unresolved. Searches for 22-product complex bilinear algorithms found no qualifying construction or matching lower bound. Fewer additions at rank 23 and finite-field lower bounds do not settle this complex-field minimum.